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9. It can be done with one weighing, with the following configuration: First, label each pile 1 through 10. Then put on the scale the same amount of coins from each pile as its label - that is, 1 from the "1" pile, 2 from the "2" pile, and so on. Now let's call the normal weight of a single real coin x. If none of the coins here were fake, the total weight would be $x+2x+3x\ldots+10x = x(1+2+3\ldots+10) = 55x$. However, knowing that exactly one full pile is fake, and weighs 1g more per coin, then if the "1" pile is fake, the total weight will be $55x+1$, the "2" pile: $55x+2$ and so on. So simply weigh the coins in this configuration to get the weight W, and $W-55x$ gives the excess weight which is the same as the label of the fake pile of coins.

As a demonstration, say a real coin weighs 10g, and the 6th pile is fake. The total weight of this configuration of coins is ~~5806, 556, 550, 550-550=6, and we know the 6th pile is fake~~

$10+20+30+40+50+66+70+80+90+100$

$= 556$

$556 - 550 = 6$